Mapping a Decade of Research at IIT Kharagpur and IIT Roorkee (2016-2026)

Assignment 2, Dynamical Processes in Complex Networks

Team 3
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***Abstract*—We collected the science and engineering papers of IIT Kharagpur and IIT Roorkee from 2016 to 2026 and linked them to 1,410 faculty members. We study them at the institution, department and faculty level, and we build a citation network and a co-authorship network. Both IITs almost doubled their output, and most of their journal papers are in Q1 journals. Their faculty h-index is about half of a career h-index, because we count only ten years of papers. Citations are heavy-tailed and grow by preferential attachment, but 88% of the citations inside an institute are self-citations, so we also rank papers without them. Faculty work in tight, hierarchical groups inside their own IIT. The collaboration network is a small world that grew denser every year, and it splits into its two campuses when about 100 bridging faculty are removed.**

I. Introduction

Throughout this course, we've looked at how networks can describe real systems, and how measures like centrality and communities tell us about their structure. In this assignment, we use these ideas on the research of two IITs, IIT Kharagpur and IIT Roorkee, over the last ten years. We look at this research at three levels, the **institution**, its **departments** and its **faculty**. Simple counts of papers and citations describe each of these levels well, but they do not tell us how the research is *connected*. For that, we need networks. A **citation network** lets us see how ideas move from one paper to another, while a **co-authorship network** lets us see which people actually work together.

We first look at the numbers for each institute as a whole, and then move down to its departments and faculty. Once we have this picture, we build the two networks and study them, and finally we see how both IITs connect with other institutes across India. The main numbers of our dataset are summarised in Table I.

TABLE I
The dataset at a glance

	IIT Kharagpur	**IIT Roorkee**
Papers, 2016-2026	25,950	20,326
Papers in 2016 $\rightarrow$ 2025	1,583 $\rightarrow$ 2,914	1,186 $\rightarrow$ 2,575
Journal papers	19,003	14,966
Q1 share of ranked journal papers	66.0%	60.7%
Journal papers with IF > 5	36.0%	34.1%
CORE A* / A conference papers	98 / 80	11 / 34
Faculty with at least one paper	770	640
Faculty h-index, mean $\pm$ SD (median)	9.6 $\pm$ 8.6 (8)	8.7 $\pm$ 7.6 (7)

II. Data and Methodology

Collecting the papers. We scraped publication records from public sources like Crossref, Europe PMC, DOAJ, INSPIRE, ORCID, DBLP and Semantic Scholar. We searched in two ways. The first way uses the institute name in the author affiliations. The second way uses the faculty list of each IIT. This gave 33,393 records for KGP and 25,775 records for Roorkee.

Filling in missing values. A lot of the scraped records had fields missing, so we filled them in using OpenAlex. This gave us the DOI, type, venue, ISSN, citations, references, author affiliations and the research topic of each paper. For journal quality, we matched each journal to its **Scimago** quartile using the ISSN and the year of publication, and for conferences we used the **CORE** rank. Later on, we went back to OpenAlex for the yearly citation counts of each paper, the impact factor of each journal, and the career profile of each faculty member. It is important to note that we did not guess any value. If something could not be found, we simply left it blank.

Checking affiliations. We kept a paper only if the IIT is in the author affiliations on the paper. This removes namesakes and papers from before a person joined the IIT.

Scope. We keep only **science and engineering**, as the professor asked. A paper is kept if its OpenAlex topic is in the Physical Sciences or Life Sciences. Health Sciences and Social Sciences papers are left out. We also leave out the faculty of humanities, management, law, education and medical units (for example HSS, the Vinod Gupta School of Management, the School of Medical Science and Technology and the B C Roy hospital). Their science papers stay though, and thus count for the institute. This scope removed 4,069 papers at KGP and 2,518 at Roorkee. We are left with **25,950** papers for KGP and **20,326** for Roorkee.

Linking papers to faculty. The handout asks for one row per paper and faculty member. To link them, we used five rules in order of reliability, the ORCID, the faculty profile we found the paper through, the exact name, the name with the department, and finally the OpenAlex author ID. A few extra checks stop wrong matches, like two faculty sharing a common surname. In the end, 82.8% of KGP papers and 84.4% of Roorkee papers have at least one faculty member, and the rest are credited to the institute.

Deliverables. The four files have 54,826 rows

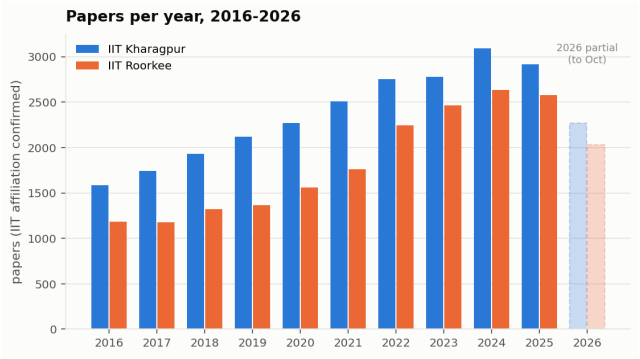


Fig. 1. Papers per year. The faded bars are 2026, which is a partial year.

(`group3_data`), 489,869 citation edges, 137,473 author pairs and 5,367 Indian partners. `group3_data` also has the number of authors of each paper and a flag for papers with more than 10 authors.

Note. 2026 is a **partial year**, and the citation counts are from October 2026. The h-index and i10-index use only the 2016-2026 papers in our dataset (Section IV-C shows what this means).

III. Institution-Level Statistics

A. Publication trend

The simplest measure of research activity is the number of papers per year. We look at it first because it sets the scale for all other results. More papers give more citations and larger networks. Thus we must keep the size of each IIT in mind when we compare them.

Both IITs grew steadily (Fig. 1). KGP went from 1,583 papers in 2016 to 2,914 in 2025, which is **1.8 times** more. Its peak was 3,094 papers in 2024. Roorkee went from 1,186 papers to 2,575, which is **2.2 times** more. KGP is larger every year, but Roorkee grows faster. Not all paper types grew at the same rate. For KGP, conference papers grew 2.4 times and journal papers grew 1.6 times. For Roorkee the values are 2.9 and 1.9. Preprints went from almost zero to 130 to 200 each year. So the growth is not only about how much the IITs publish, but also about *how* they publish. The 2026 bars are low simply because we only have data for a few months.

B. Journal quality

Counting papers tells us nothing about their quality. For this, we use the Scimago quartile, which ranks the journals of each subject by their SJR score and splits them into four equal parts, with Q1 being the top 25%. Since journals are ranked *within* their own subject, the Q1 share lets us compare fields that cite at very different rates.

About 95% of the journal papers of both IITs have a quartile. Of these, 66.0% at KGP and 60.7% at Roorkee are in Q1 (Fig. 2), between 27% and 30% are in Q2, and only about 1% are in Q4. This is what we would expect from two of the oldest IITs, which mostly publish in well established journals.

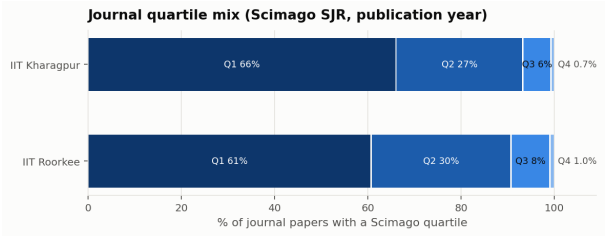


Fig. 2. Journal quartile mix (Scimago, year of publication).

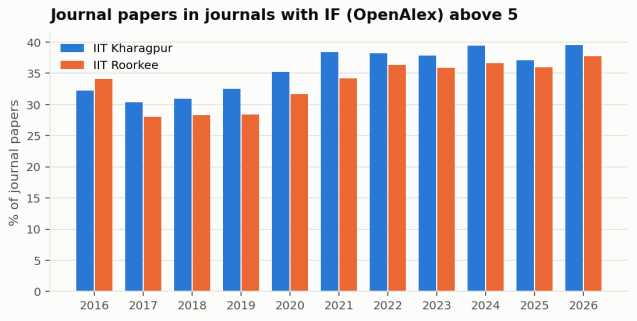


Fig. 3. Share of journal papers in journals with IF (OpenAlex) above 5, per year.

C. Impact factor

While the quartile only places a journal in one of four bands, the impact factor (IF) gives us the number of citations a journal gets in a year to its papers from the two years before, divided by the number of those papers. Clarivate's official Journal Impact Factor is not free, so we apply the same formula to OpenAlex data, which OpenAlex calls the *2-year mean citedness*. It differs from the official IF in two ways. It is the current value, not the one from the year of publication, and it usually runs higher since OpenAlex counts more citing sources. For example, Scientific Reports has 5.2 here but about 3.8 in Clarivate, so the thresholds below are on the OpenAlex scale.

The median journal paper sits in a journal with an IF of 4.1 at KGP and 4.0 at Roorkee. 36.0% of KGP journal papers and 34.1% of Roorkee journal papers are in journals with IF > 5, and 6.1% and 5.3% are in journals with IF > 10 (1,148 and 797 papers). The share above 5 rose a little over the decade, from about 30% in 2017-2019 to between 36% and 39% since 2021 (Fig. 3). This agrees with the Q1 share above, with KGP slightly ahead on both.

D. Top journals and important journals

Table II shows the top 10 journals of each IIT with their IF. All are Q1, except ChemistrySelect (Q2). Some are broad journals, such as Scientific Reports and RSC Advances. Others match a strong area of each IIT, such as fluid mechanics at KGP (Physics of Fluids) and nuclear physics at Roorkee (Physical Review C).

Some journal groups are seen as important in physics and chemistry. These are the American Physical Society (APS)

TABLE II
TOP 10 JOURNALS OF EACH IIT BY PAPERS, WITH IF (OPENALEX)

Journal	Papers	IF
IIT Kharagpur		
Physics of Fluids	325	4.2
Scientific Reports	175	5.2
ACS Applied Materials & Interfaces	149	8.8
Langmuir	130	4.9
RSC Advances	112	7.5
Physical Chemistry Chemical Physics	107	3.6
Journal of Alloys and Compounds	96	6.3
ACS Applied Nano Materials	95	6.9
Science of The Total Environment	93	12.2
Materials Science and Engineering A	88	6.8
IIT Roorkee		
Scientific Reports	124	5.2
Int. J. of Biological Macromolecules	120	9.2
Physical Review C	117	3.2
IEEE Trans. on Industry Applications	110	4.4
Environ. Science and Pollution Research	101	6.4
Physics of Fluids	86	4.2
IEEE Access	85	6.4
ChemistrySelect	78	2.5
IEEE Trans. on Electron Devices	77	3.6
J. of Environmental Chemical Engineering	70	6.1

TABLE III
PAPERS IN IMPORTANT JOURNAL GROUPS, 2016-2026

Group	KGP	Roorkee
APS (all Physical Review journals)	299	231
of which PRL	10	6
SPIE journals	17	22
SPIE conference proceedings	8	8
Nature journals (Nature, Nature Communications, ...)	25	22
JACS	9	14

journals (the Physical Review family), Physical Review Letters (PRL), the SPIE journals, the Nature journals and the Journal of the American Chemical Society (JACS). Counting these papers shows how often the IITs publish at the top of these fields. The counts are in Table III. The APS journals are the largest group, with about 300 papers at KGP and 230 at Roorkee. At KGP they are spread over Physical Review A, B, D and E, and at Roorkee half of them are in Physical Review C (nuclear physics). Papers in PRL, Nature journals and JACS are rare, with 6 to 25 papers each over ten years. This is expected, because these journals accept very few papers.

E. Conferences and CORE

For conference papers we use the **CORE** rank (A*, A, B, C). CORE ranks mostly computing conferences. Thus it tells us about the computer science side of each IIT, and most conference papers have no CORE rank.

At KGP, 522 of 3,814 conference papers (13.7%) are in CORE-ranked venues. These are 98 A*, 80 A, 222 B and 122 C papers. At Roorkee, 289 of 3,211 (9.0%) are ranked. These are 11 A*, 34 A, 144 B and 100 C papers. Thus KGP is far ahead at the top ranks. Computer Science is 14.1% of KGP papers and only 8.9% of Roorkee papers. The most

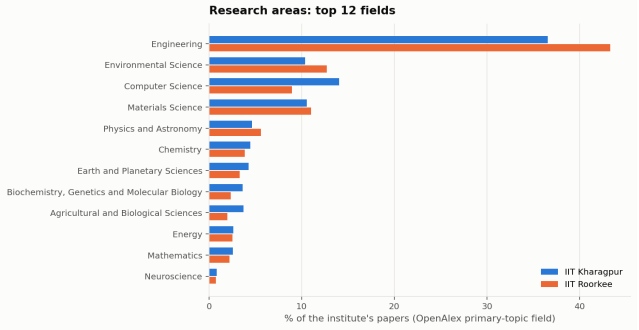


Fig. 4. Share of papers by OpenAlex field (primary topic).

frequent CORE venue is the same for both IITs. It is the IEEE Geoscience and Remote Sensing Symposium (rank C, 51 KGP papers and 81 Roorkee papers). The KGP list also has top venues such as AAAI (A*, 16 papers) and CIKM (A, 17 papers). Ranked conference papers increased after 2021 at both IITs.

F. Research areas and their categories

Next, we look at what each IIT works on. OpenAlex places every paper in a four-level hierarchy, the *domain*, *field*, *subfield* and *topic*. For example, a paper on perovskite solar cells falls under the domain Physical Sciences, the field Engineering, the subfield Electrical and Electronic Engineering, and the topic Perovskite Materials and Applications. These research areas matter for the rest of the report too, since they decide which papers can cite each other and which faculty can work together.

Engineering is the largest field at both IITs, with 36.6% of KGP papers and 43.3% of Roorkee papers (Fig. 4). Inside Engineering the main subfields are Electrical and Electronic Engineering (24% at KGP and 32% at Roorkee), Mechanical Engineering, Biomedical Engineering and Civil and Structural Engineering. After Engineering, KGP has more Computer Science (14.1%) and Roorkee has more Environmental Science (12.7%). Materials Science is about 11% at both. The dashboard lists the subfields inside every field.

Why is Environmental Science so high? It is 10.4% of KGP papers and 12.7% of Roorkee papers. The subfields explain it (Fig. 5). About two thirds of these papers are in Environmental Engineering, Water Science and Technology, and Global and Planetary Change. The most common topics are hydrology, watershed management, floods, climate models and the removal of pollutants from water. The departments of the authors are mostly engineering departments. At KGP these are Civil Engineering (28% of these papers), Agricultural and Food Engineering and the CORAL centre. At Roorkee they are Civil Engineering (21%), Hydrology and Water Resources Development and Management. Thus Environmental Science is high, because OpenAlex puts much of the water and climate engineering of the two IITs in this field. Roorkee has entire departments for hydrology and water resources, so its share is higher.

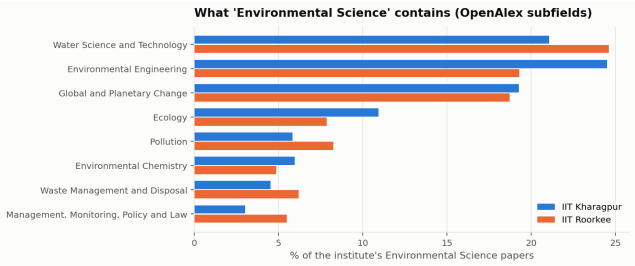


Fig. 5. Subfields inside Environmental Science.

TABLE IV
MOST-CITED PAPERS (WORLDWIDE, OCTOBER 2026)

	Paper (year)	Cit.
KGP	Soil carbon 4 per mille (2017)	2,020
KGP	SymPy, symbolic computing in Python (2017)	1,502
KGP	iFogSim, a toolkit for modeling and simulation of resource management techniques ... (2017)	1,250
Roork.	The second data release from the European Pulsar Timing Array (2023)	1,069
Roork.	A comprehensive review on NSGA-II for multi-objective combinatorial optimization (2021)	971
Roork.	Microwave–material interaction phenomena, heating mechanisms ... (2016)	876

G. Most-cited papers and large collaborations

The last measure at this level is the number of worldwide citations of each paper. The most cited papers are the most visible work of each IIT, but some of them say more about the *type* of paper than about the institute itself.

Table IV shows the top three papers of each IIT. At KGP, two of them are software tools (SymPy and iFogSim). At Roorkee, the top paper is a large collaboration (the European Pulsar Timing Array), and at least five of the top ten are reviews or surveys. This makes sense, since anyone working on a topic or using a tool ends up citing its review or tool paper, so such papers collect citations very quickly.

We **flag papers with more than 10 authors**, because in such papers our faculty are one group among many. They are few, 534 papers at KGP (2.1%) and 454 at Roorkee (2.2%). 17 and 7 of them have 100 or more authors. But they get more than their share of citations, 6.0% of all KGP citations and 3.7% of all Roorkee citations. The flag is in `group3_data.csv` and on the dashboard.

H. How much do large collaborations matter?

To see how much our numbers depend on these papers, we recomputed every headline number without them (Fig. 6, left). The *typical* paper barely changes. The median paper still has 6 citations at KGP and 5 at Roorkee, and the Q1 share drops by only half a point. The *top*, however, does change. At KGP, 6 of the 10 most cited papers have more than 10 authors (2 at Roorkee). Without them, KGP loses 9.3% of its papers with 100 or more citations, and its institute h-index falls from 184 to 176, while Roorkee's falls from 156 to 154. The faculty h-index moves too, with 74 KGP faculty (9.6%) and 57 Roorkee faculty (8.9%) losing at least one point, and the median KGP faculty h-index dropping from 8 to 7.

A citation premium. Large papers get more citations than other papers from the same IIT and year (Fig. 6, right). At KGP, the median large paper gets 2.7 times the citations of the median other paper (median over 2016-2024), while at Roorkee the premium is smaller, at 1.25 times. One reason is simply that more authors means more people citing the paper, and more networks that come across it.

What these papers are. 82% of the large journal papers at KGP are in Q1 journals, against 62% for its other papers (Roorkee 76% against 57%). Only 2-3% are conference papers, against 15-16% of the other papers, and 81% have a co-author abroad (Roorkee 76%), against 22-23% of the rest. So large teams mostly mean international collaboration. Most of them are not huge either, as 86% of the KGP and 76% of the Roorkee large papers have 11 to 20 authors, and only 17 and 7 papers have 100 or more.

Where they come from. At KGP, they mostly come from materials research, with Metallurgical and Materials Engineering (78 papers), Physics (60), Agricultural and Food Engineering (60) and Biosciences (45). Materials Science is the largest field (22% of them), and the top venues are *Small*, *Advanced Functional Materials* and *ACS Applied Materials & Interfaces*. At Roorkee, they come mostly from Physics (134 papers, 30% of Roorkee's large papers and 9% of its physics papers), followed by Biosciences (74) and Hydrology (28). The top venue here is *Physical Review C* (44 papers) from nuclear physics collaborations, and the pulsar timing papers show up too.

Who depends on them. For a few faculty, large papers make up a big part of their h-index. At KGP, the h-index of Dr. Tapas Kumar Maiti (Biosciences) falls from 36 to 30 without them, and that of Dr. Abhijit Mukherjee (Geology and Geophysics) from 44 to 39. At Roorkee, the largest drop is in Physics, where Dr. Bhal Chandra Joshi, who works on pulsar timing arrays, goes from 8 to 2. This does not make these papers any less valuable. But an h-index built on large collaborations measures something different from one built in a small group, which is why we show both on the dashboard.

IV. FACULTY AND DEPARTMENT LEVEL

Everything so far is really a sum over people. So we now look at the faculty themselves, and then at the departments they belong to.

A. h-index statistics

A faculty member has an **h-index** of h if h of their papers have at least h citations each. This measure combines how much a person publishes and how much other people cite it. One very large paper cannot increase it alone.

The h-index is strongly right-skewed (Fig. 7). Many faculty have a small h-index, and a few have a very large one. Thus

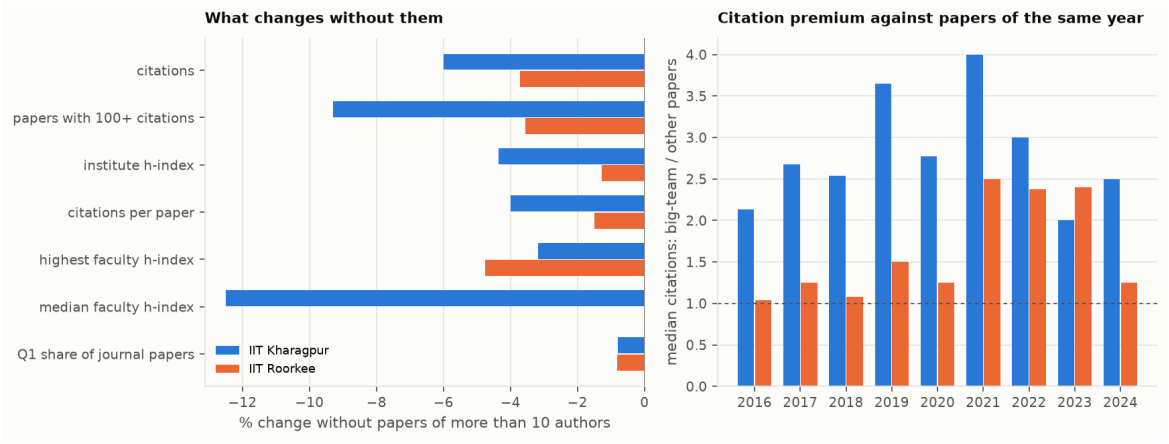


Fig. 6. Papers with more than 10 authors. Left: change of the headline numbers without them. Right: their median citations divided by the median of the institute's other papers of the same year.

TABLE V
FACULTY H-INDEX AND I10-INDEX (2016-2026 PAPERS)

	h-index		i10-index	
	KGP	Roorkee	KGP	Roorkee
Faculty	770	640	770	640
Mean ± SD	9.6 ± 8.6	8.7 ± 7.6	13.9 ± 21.7	11.6 ± 17.2
Median	8	7	6	5
Middle half (Q1–Q3)	3–13	3–12	1–16	1–14
90th percentile	22	19	37	30
Maximum	63	42	184	122

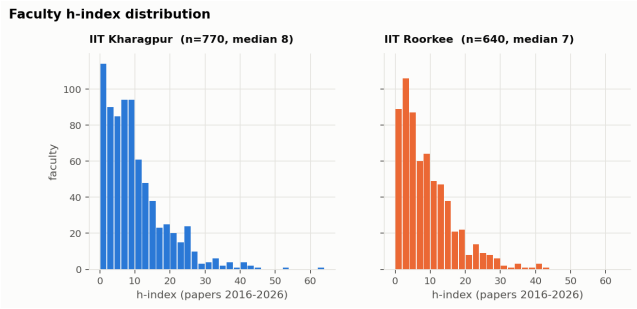


Fig. 7. Distribution of the faculty h-index (2016–2026 papers in our dataset).

the mean alone is misleading, and we give several statistics (Table V). At KGP the mean is 9.6 with a standard deviation of 8.6, and the median is 8. At Roorkee the mean is 8.7 with a standard deviation of 7.6, and the median is 7. The SD is almost as large as the mean, which again shows the long tail. The highest values are 63 at KGP (Dr. Narayan Chandra Das, Rubber Technology) and 53 (Dr. Makarand M. Ghangrekar, Civil Engineering), and 42 at Roorkee (Dr. Soumitra Satapathi, Physics).

B. i10-index

The i10-index is the number of papers with at least 10 citations. It is used by Google Scholar. It is simpler than the h-index, and it shows how many of a person's papers are cited

TABLE VI
MEDIAN FACULTY H-INDEX, THREE WAYS

	KGP	Roorkee
Our h-index (2016–2026 papers)	9	9
Same papers, OpenAlex citation counts	10	9
OpenAlex profile, whole career	20	19

well. The median faculty member has 6 such papers at KGP and 5 at Roorkee in these ten years. The mean is much higher (13.9 and 11.6), because a few very active labs have more than 100 such papers each (Table V).

C. Why is the computed h-index low?

A median h-index of 7 or 8 seems low for faculty at two of the oldest IITs, so we looked into why. For each faculty member, we compare three values, the h-index we computed, the h-index of the same papers using OpenAlex citation counts instead, and the h-index of their whole OpenAlex profile, which covers their entire career. We only use the career value when the profile clearly belongs to that person (at least 5 of their papers checked, with at least 80% of them on that profile), since some profiles merge two people with the same name. This leaves 565 faculty at KGP and 437 at Roorkee.

Table VI gives the answer. The citation source barely matters, as the same papers give almost the same h-index with OpenAlex counts. The real difference is the *time window*. The computed h-index only counts papers from 2016 to 2026, while a career h-index also counts all the older papers, which have had many more years to collect citations. So the computed value is about half of the career value (53% at KGP and 50% at Roorkee at the median), and 98% of faculty are below their career value (Fig. 8). Google Scholar is higher still, since it also counts preprints, theses and books. For example, Dr. A. Taraphder's 2017 PRL paper has 351 citations in our data and 435 on Google Scholar, and the paper is in our dataset, linked to him by his ORCID.

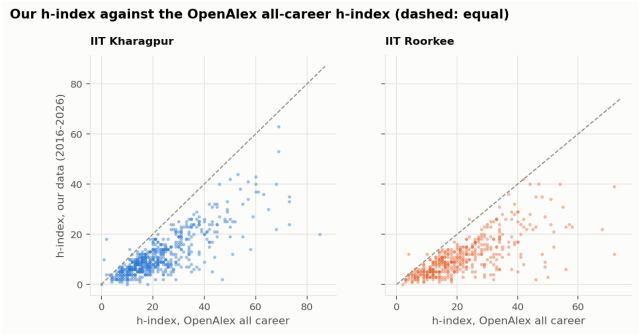


Fig. 8. Our h-index against the OpenAlex whole-career h-index (dashed line: equal).

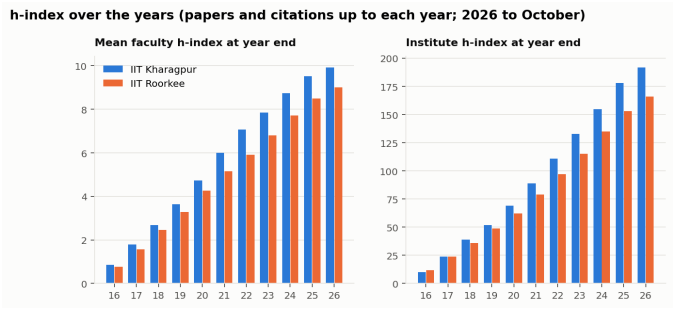


Fig. 9. h-index at the end of each year. Left, the mean faculty h-index. Right, the h-index of each institute.

D. h-index over the years

The h-index grows with time, because a person writes more papers and the old papers collect citations. To see this growth, we compute the h-index at the end of each year. For year Y we use the papers published up to Y and only the citations received up to Y, from the yearly citation counts in OpenAlex.

The mean faculty h-index rose from 0.9 at the end of 2016 to 9.5 at the end of 2025 at KGP, and from 0.8 to 8.5 at Roorkee (Fig. 9). It grows by about one point each year at both IITs, and KGP stays a little ahead. The h-index of each institute as a whole, computed over all its papers, grew from 10 to 178 at KGP and from 12 to 153 at Roorkee. The growth is steady. This agrees with the growing output and the growing citations from Indian institutes (Section V).

E. Departments

Departments are between the person and the institute. They are the natural unit to compare research areas inside one IIT. Table VII shows the five departments with the most papers at each IIT.

At KGP, Computer Science has the most papers, but Civil Engineering has the most citations (41,228). Civil Engineering includes environmental engineering, and two of the five highest h-index values at KGP are in this department. At Roorkee, Civil Engineering has the most papers and the most faculty (75). The Q1 share is between 50% and 73% in the engineering departments. Mathematics is much lower at both IITs, at about 45%. We expect this, because mathematics journals have lower impact scores in general. The mean h-index of a department is close to its SD, so every department has a mix of young and very senior faculty.

TABLE VII
TOP 5 DEPARTMENTS OF EACH IIT BY NUMBER OF PAPERS

Department	Fac.	Papers	Cit.	Q1	h, mean±SD
IIT Kharagpur					
Computer Sci. & Engg.	51	2,429	25,758	66.9%	9.0 ± 7.9
Mechanical Engg.	54	1,926	29,048	71.3%	11.0 ± 9.3
Civil Engg.	45	1,748	41,228	70.8%	12.7 ± 10.9
Chemistry	42	1,650	36,372	72.8%	14.0 ± 9.3
Agri. & Food Engg.	52	1,480	32,085	61.5%	10.8 ± 8.5
IIT Roorkee					
Civil Engg.	75	2,021	26,979	56.2%	8.5 ± 6.6
Electronics & Comm.	45	1,854	15,978	49.7%	7.9 ± 6.0
Electrical Engg.	48	1,694	16,424	68.7%	8.3 ± 6.6
Mech. & Industrial Engg.	53	1,671	30,210	61.4%	11.2 ± 8.2
Physics	44	1,515	20,711	65.8%	9.4 ± 7.9

V. Citation Network

Up to now, all citations had the same value. A network lets us ask *who* cites whom.

A. Construction

Each node is a paper of our two IITs. A directed edge $A \to B$ means that paper A cites paper B, and both papers are in our dataset. The edges have a direction because a citation goes one way, from a newer paper to an older paper. The network has 46,072 papers and 86,090 edges. KGP has 25,942 papers and 47,586 edges, and Roorkee has 20,326 papers and 35,246 edges. The network is very sparse. The density is 4.1×10^{-5} and the mean in-degree is 1.9. About half of the papers get no citation from another paper of the two IITs. The largest weakly connected component has 53% of the papers. We expect this, because the papers are in many fields that do not cite each other.

B. In-degree distribution

As we know from our lectures already, the in-degree of a node is the number of edges that point to it. Here it is the number of papers in our dataset that cite a paper. Its distribution shows if attention is spread evenly or goes to a small number of papers.

Fig. 10 shows $P(\text{in-degree} \geq k)$ on a log-log scale. The curve decreases slowly over two orders of magnitude. Thus the distribution is heavy-tailed. Most papers get zero, one or two internal citations, but a few papers get 40 to 58. The curves of the two IITs are almost the same. This tells us that the shape comes from how citation works, and not from one institute.

C. Is it a power law?

A straight line on a log-log plot is not enough to call a distribution a power law. We follow the method of Clauset, Shalizi and Newman. It fits $P(k) \propto k^{-\alpha}$ above a cut-off $x_{\min}$ by maximum likelihood. Then it compares the power

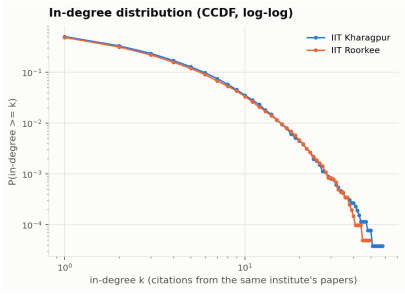


Fig. 10. In-degree distribution of the citation network (CCDF, log-log).

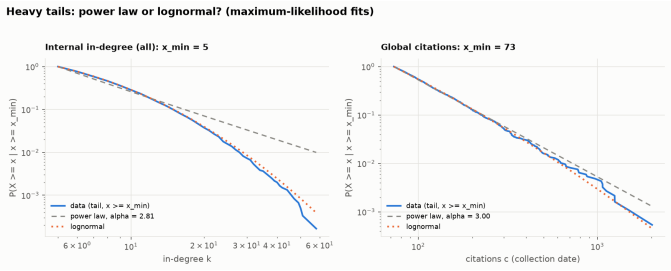


Fig. 11. Maximum-likelihood fits of the tail: power law (grey) and lognormal (orange).

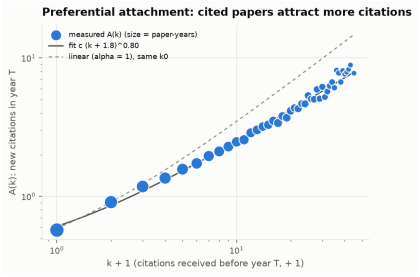


Fig. 12. Attachment rate $A(k)$: new citations in a year for a paper with k earlier citations.

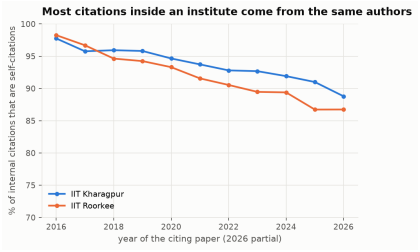


Fig. 13. Share of internal citations that are self-citations, by year of the citing paper.

law with other heavy-tailed distributions by a likelihood ratio test (Fig. 11).

For the internal in-degree, the fit gives $\alpha = 2.81$ above $x_{\min} = 5$. But the lognormal fits much better (normalised ratio $R = -13.3$, $p < 0.001$), because the curve bends down at the end. This is natural. A paper can get at most a few dozen citations from inside two institutes. For the global citation counts, the tail above 73 citations (1,861 papers) has $\alpha = 3.00 \pm 0.05$. The power law is much better than an exponential ($R = 5.4$, $p < 0.001$), but we cannot tell it apart from a lognormal ($p = 0.16$). Thus we can say "heavy-tailed", but not "scale-free". This is also what large studies of citations find.

D. Preferential attachment, measured

In the lectures, the Barabási–Albert model explains heavy tails with preferential attachment: a node with degree k gets new links at a rate $\Pi(k) \propto k$. With dated citations we can measure this directly. For each year T from 2017 to 2025, we take every paper that already existed, count its citations k up to year $T-1$, and count its new citations in year T. The mean number of new citations at each k is the attachment rate $A(k)$. We use the internal citations and the citations from other Indian institutes, because both have a year.

Fig. 12 shows that $A(k)$ grows steadily with k. A paper with 10 citations gets about three times as many new citations as a paper with one. The best fit is $A(k) = c\,(k + k_0)^{\alpha}$ with $\alpha = 0.80 \pm 0.02$ and $k_0 = 1.8$. Thus the attachment is slightly sublinear. The k_0 is the "initial attractiveness": an uncited paper still gets 0.57 new citations a year. A sublinear α gives a stretched-exponential or lognormal shape rather than a pure power law. This agrees with the fits above. One caution: papers also differ in quality (their "fitness"), and better papers are both cited earlier and cited later. Part of the measured effect can come from this.

E. Self-citations

The brief defines the in-degree as the number of citations from the same institute's papers. We checked who these citations come from. An edge is a self-citation if the citing and the cited paper share an author (the same OpenAlex author, or the same faculty member). 88.5% of the 86,090 internal edges are self-citations (KGP 92.8%, Roorkee 90.1%). Only 9,909 edges connect different authors. The share falls from 98% in 2016 to 87–89% in 2026 (Fig. 13). Over time, more colleagues have papers that one can cite.

This changes how we must read the internal network. It mostly shows how researchers build on *their own* earlier work. 42 papers have zero citations in the world but five or more internal citations, and all of these are self-citations. 79% of the edges stay inside one OpenAlex field (field assortativity 0.73), and 76% of the non-self edges do too. Thus ideas move between fields only rarely, even between colleagues.

F. PageRank

As we know from our lectures already, **PageRank** gives a score to a node. The score depends on how many nodes point to it *and* on how important those nodes are.

$$PR(i) = \frac{1-\alpha}{N} + \alpha \sum_{j \to i} \frac{PR(j)}{k_{\text{out}}(j)}, \qquad \alpha = 0.85$$

Here N is the number of papers and $k_{\text{out}}(j)$ is the number of papers that j cites. The in-degree gives the same value to

TABLE VIII
TOP 5 PAPERS BY PAGERANK (BOTH IITS), WITH IN-DEGREE AND IN-DEGREE RANK

	Paper (year)	In	Rank
KGP	Hydrothermal growth of few layer 2H-MoS_2 for heterojunction ... (2016)	20	195
KGP	Uniform electric-field-induced lateral migration of a sedimenting drop (2016)	58	1
KGP	V_2O_5 microflower decorated cathode for enhancing power ... (2016)	23	108
Roork.	A review of power electronic converters for variable speed pumped storage ... (2018)	49	4
KGP	Groundwater quality and depletion in the Indo-Gangetic Basin ... (2016)	51	2

every citation. PageRank gives more value to a citation from an influential paper. Thus it finds papers that are the *base* of other well-cited work.

Our data shows this clearly (Table VIII). The top paper by PageRank is a 2016 KGP paper on MoS_2 heterojunctions. It has only 20 internal citations and its in-degree rank is 195. But the papers that cite it have many citations themselves. There is also an age bias. Seven of the top ten papers are from 2016, and the Spearman correlation between PageRank and publication year is -0.39. This makes sense. In a citation network the score moves almost only from newer papers to older papers. Thus the oldest papers in our window collect the most score.

G. Correcting PageRank

The two problems above, self-citations and age, also affect PageRank. We made two corrections.

First, we ran PageRank on the 9,909 non-self edges. Only 7 of the top 100 papers stay in the top 100. The new top papers are cited by other groups. Examples are a review of Himalayan–Karakoram glaciers (Roorkee, 331 citations in the world) and studies of rainfall and drought over India (KGP). Second, we removed the age bias with the rescaled PageRank of Mariani, Medo and Zhang. Each paper's score becomes a z-score among the papers of the same year: $R_i = (p_i - \mu_{y(i)})/\sigma_{y(i)}$. The test of a fair ranking is simple. Every year should have about 1% of its papers in the overall top 1%. Fig. 14 shows the result. With plain PageRank, 6.5% of the 2016 papers are in the top 1%, but no paper from 2024 or 2025. The coefficient of variation of these shares falls from 1.44 (PageRank) to 0.87 (no self-citations) and to 0.20 (rescaled). In the rescaled top 10, papers from 2023–2025 appear next to older ones. These are the papers that started strongly for their age.

H. Citation flow with other Indian institutes

The network above stays inside our two IITs. To see their place in Indian research, we also count citations to and from other Indian institutes. The in-flow is the number of citations from papers of other Indian institutes to our papers. The out-flow is the number of citations from our papers to their papers.

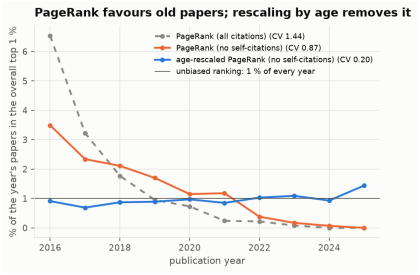


Fig. 14. Share of each year's papers in the top 1% of three rankings (2016–2025 papers).

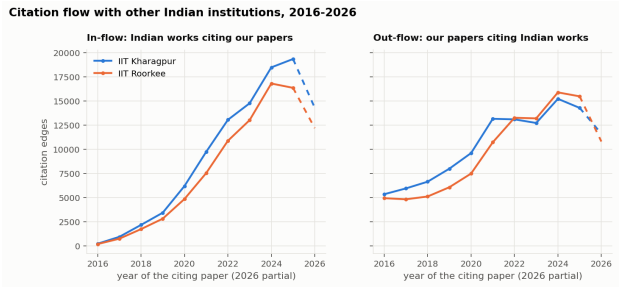


Fig. 15. Citation in-flow and out-flow with other Indian institutes, per year.

The impact ratio is the in-flow divided by the out-flow. A ratio above 1 means that Indian research cites the IIT more than the IIT cites Indian research.

KGP has an in-flow of 102,511 and an out-flow of 115,479, so its ratio is 0.89. Roorkee has 86,983 and 107,630, so its ratio is 0.81. At first, it seems that both IITs give more citations than they get. Fig. 15 shows the reason. In 2016 our papers already cited thousands of older Indian papers (the KGP out-flow was 5,335). But almost no paper had cited them yet (the KGP in-flow was 214), because citations take years to arrive. The in-flow grows much faster over time. It is larger than the out-flow since 2023 at KGP and since 2024 at Roorkee. Thus the total ratio below 1 comes mostly from the 2016–2026 window.

The largest Indian source of citations to KGP is IIT Roorkee (2,914 citations), followed by VIT and AcSIR. For Roorkee the largest sources are UPES (2,345), VIT and AcSIR. Large universities such as VIT, UPES and Saveetha cite our IITs much more than our IITs cite them (ratios between 1.6 and 3.7). The most cited Indian paper that cites KGP is the SciPy 1.0 paper, and the most cited one that cites Roorkee is a paper on drug delivery vehicles for cancer treatment.

VI. CO-AUTHORSHIP NETWORK

The citation network shows how *ideas* move. The co-authorship network shows how *people* work together. This is closer to how research groups really work.

A. Construction

Each node is a faculty member of one of the two IITs. An undirected edge connects two faculty who wrote at least one paper together. The edge weight is the number of joint papers.

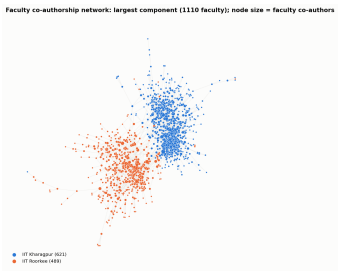


Fig. 16. Largest component of the faculty network. KGP is blue and Roorkee is orange.

The network has 1,410 faculty and 2,719 edges, and its density is 0.0027. On average, a faculty member has 3.9 faculty co-authors, and 254 faculty have no faculty co-author. The largest component has 79% of all faculty. Fig. 16 shows it. The two IITs make two separate clusters, and only 77 edges connect them. This is natural, because the two campuses are far apart and most work happens inside one campus.

The faculty network is only a small part of all co-authorship. A faculty member has a median of 25 distinct co-authors. Of all the author pairs in `authorship_network.csv` with a faculty member, 52% are with students and staff of the same IIT, 21% with other Indian institutes, 17% with foreign institutes, and only 7% with another faculty member of our two IITs. Thus the faculty network shows the links *between* research groups, and the students fill the groups.

B. Degree centrality

As we know from our lectures already, **degree centrality** is the number of neighbours of a node, divided by $n-1$. In this network it is the number of faculty colleagues that a person wrote papers with. Thus it finds the people who collaborate the most.

The highest values are for Dr. Partha Roy (Roorkee, Biosciences and Bioengineering, 41 faculty co-authors), Dr. Suman Chakraborty (KGP, Mechanical Engineering, 40) and Dr. Chandra Sekhar Tiwary (KGP, Metallurgical and Materials Engineering, 36). Work in these fields often needs equipment and skills from several departments, so we expect a high degree here.

C. Betweenness centrality

As we know from our lectures already, **betweenness centrality** measures how often a node is on the shortest path between two other nodes. A person with a high betweenness is a bridge. Without this person, some groups are far apart or not connected. This role is different from having many co-authors.

The top bridges are Dr. Partha Roy (0.054) and Dr. Pradip K. Maji (0.038) at Roorkee, and Dr. Ramkrishna Sen (0.035) and Dr. Narayan Chandra Das (0.031) at KGP. Dr. Pradip K. Maji shows the difference between the two measures. He has only 20 faculty co-authors, half as many as the top people by degree, but his co-authors are in different groups. Polymer and process engineering is used by chemists, materials scientists and engineers, so it makes sense that he connects them.

Do central faculty have more impact? Degree and h-index are strongly correlated (Spearman $\rho = 0.70$; betweenness 0.67). But central faculty also write more papers, and more papers give a higher h-index. If we hold the number of papers fixed (partial rank correlation), ρ falls to 0.15 for degree and 0.08 for betweenness. Thus most of the link is productivity. Working with many colleagues helps the h-index only a little beyond the extra papers.

D. Communities

A community is a group of nodes with many edges inside the group and few edges to other groups. We use the Louvain algorithm, which finds the groups with the highest **modularity**.

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j)$$

The modularity compares the number of edges inside the groups with the number we expect at random. Here the communities are the real research groups. These groups can follow department lines, or not.

We get $Q = 0.79$ and 26 communities with 5 or more faculty. In 23 of these 26 communities, 90% or more of the members are from one IIT. This again shows that collaboration stays inside each campus. The communities are interdisciplinary. In a typical community, the largest department has only about 46% of the members. For example, the largest community has 139 faculty, almost all from Roorkee. Biosciences is its largest department, but its members come from 20 departments. The result is stable. Over 10 random seeds, the mean Adjusted Rand Index is 0.84. Fig. 17 shows the eight largest communities.

Communities and departments. How much do the communities follow the departments? The normalised mutual information between the two labels is 0.57, half-way between "unrelated" (0) and "the same" (1). 49% of the edges join two faculty of the same department, and the department assortativity is 0.47. For the institute it is 0.94. Thus the campus decides almost everything, and the department about half.

Louvain or Leiden? The brief allows both. Louvain can return a community that is not connected inside, and Leiden fixes this. We ran Leiden (seed 42) and compared. At resolution 1, Leiden finds 25 communities with 5 or more faculty and $Q = 0.786$, against 26 and $Q = 0.785$ for Louvain, and the two partitions agree well (ARI 0.75). No community of either method is disconnected. With a resolution sweep (0.25 to 4), the number of communities with 5 or more faculty goes from 8 to 64, and the large communities split into smaller groups. The structure is hierarchical: there are groups inside groups.

E. Comparison with random networks

As we know from our lectures already, the **clustering coefficient** is the probability that two co-authors of a person

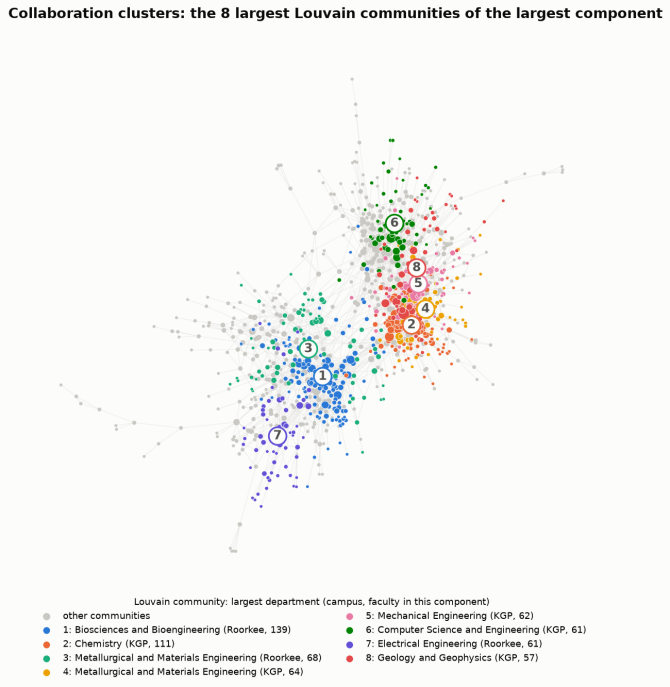


Fig. 17. Collaboration clusters: the eight largest Louvain communities in the largest component.

TABLE IX
FACULTY NETWORK AND TWO RANDOM MODELS (MEAN OF 10 SAMPLES)

Metric	Real	ER	Rewired
Clustering coefficient	0.168	0.002	0.007
Modularity (Louvain)	0.785	0.538	0.464
Degree assortativity	0.121	0.00	−0.03
Mean shortest path (giant comp.)	5.34	5.48	4.28

also wrote a paper together. A value such as 0.17 has a meaning only when we compare it with a baseline. Thus we compare the network with two random models of the same size, with 10 samples each (Table IX). The first model is an Erdős–Rényi (ER) graph with the same number of edges. The second model is a degree-preserving rewiring that keeps the degree of every node.

The real network has about 70 times more clustering than the ER graph and 23 times more than the rewired graph. But its mean shortest path is almost the same as in the ER graph. This is the small-world property. The small-world coefficient $\sigma = (C/C_{\mathrm{ER}})/(L/L_{\mathrm{ER}})$ is 72, far above 1. The network has tight local groups, but the paths between people are still short. This makes sense for research groups. A new project often brings in a colleague of a colleague. The modularity is also much higher than in the two random models. Thus the communities above are real, and the algorithm did not create them. The positive assortativity (0.12) shows that well-connected faculty usually work with other well-connected faculty.

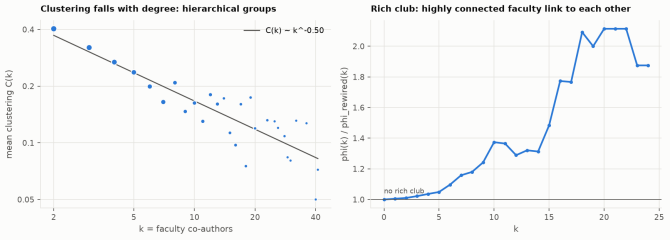


Fig. 18. Left: clustering spectrum $C(k)$. Right: rich-club coefficient divided by its rewired value.

F. Core, hierarchy and rich club

The **k-core** is the largest group in which every member has at least k co-authors inside the group. The deepest core is $k = 6$ and has 23 faculty, all at KGP, mostly in materials science, metallurgy, rubber technology, physics and biosciences. This is the dense centre of Fig. 17. The clustering coefficient falls with the degree as $C(k) \sim k^{-0.50}$ (Fig. 18, left). Faculty with few co-authors sit in one tight group, and hubs join several groups that are not linked to each other. This is the signature of a hierarchical network from our lectures. The **rich-club coefficient** $\phi(k)$ is the density of links among faculty with more than k co-authors. Divided by its value in the rewired graphs, it rises from 1 to about 2 for $k \geq 18$ (Fig. 18, right). Thus the best-connected faculty work with each other about twice as often as their degrees alone explain.

VII. Dynamics of the Collaboration Network

Our course is about *dynamical* processes. We look at three of them: how the network grew, how it breaks when people leave, and how something spreads on it.

A. Growth

We date each edge by the first paper of the pair and build the network at the end of each year (Fig. 19). In 2016 there were 312 edges among 361 faculty, and the largest component had only 11% of them. In 2017 it had 44%, and in 2018 already 74%. This is the emergence of a giant component from the lectures. It happened while the mean degree grew only from 1.7 to 2.4. Since then the largest component grew to 96%, and the mean degree grew from 1.7 to 4.7. The edges grow faster than the nodes, $E \sim N^{1.90}$. This is densification, as Leskovec and others found for many real networks. The links between the two IITs grew from 4 to 77.

B. Robustness

What happens when faculty leave or retire? We remove faculty step by step and measure the largest component (Fig. 20). For random removal, the network is robust. We must remove 41% of the faculty before the largest component halves. The mean-field theory from the lectures (Molloy–Reed, $\kappa = \langle k^2 \rangle / \langle k \rangle = 9.1$) predicts a breakdown only at $f_c = 1 - 1/(\kappa - 1) = 0.88$. For a targeted attack the network is fragile. Removing the faculty with the highest degree halves it after 14%. Removing the faculty with the

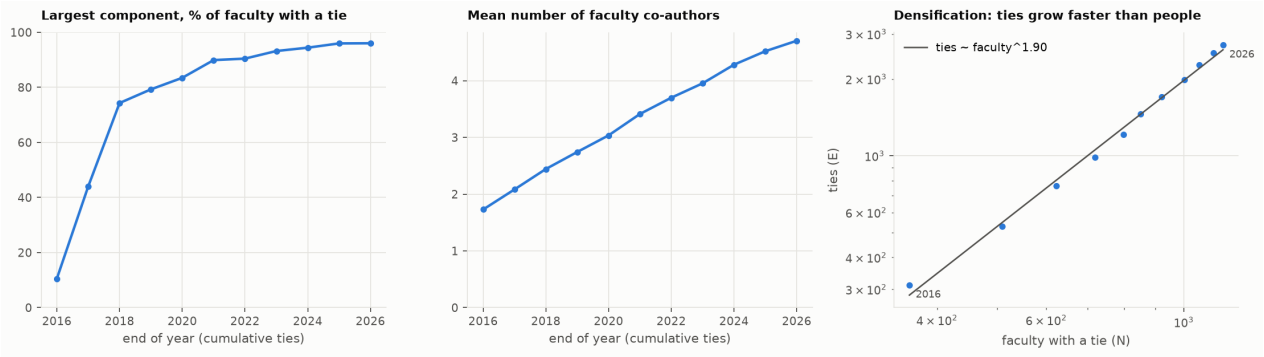


Fig. 19. Growth of the faculty network: largest component, mean degree, and densification (E against N, log-log).

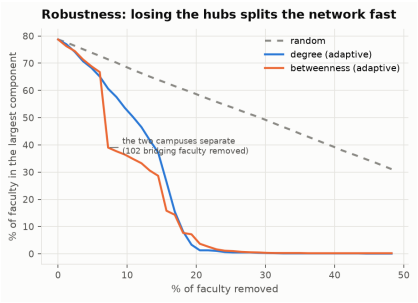


Fig. 20. Largest component as faculty are removed: random, by degree, by betweenness (both recomputed after each step).

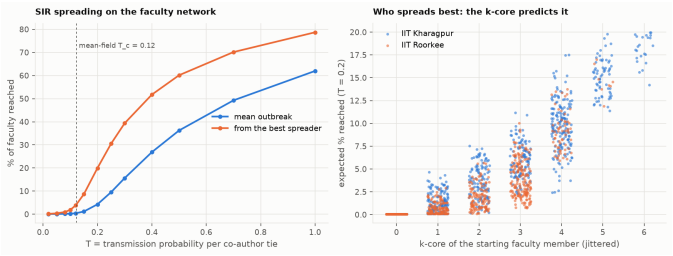


Fig. 21. SIR spreading. Left: outbreak size against T. Right: expected outbreak against the k-core of the start.

highest betweenness halves it after only 7%. At that point (102 faculty) the network falls into a Kharagpur part and a Roorkee part. This is the known weakness of heterogeneous networks: they survive random failure but not the loss of their hubs and bridges.

C. Spreading

Finally, imagine that a new idea or method spreads along the co-author links, like an **SIR** epidemic. Each "infected" faculty member passes it to each co-author with probability T, and then stops. This model is exactly bond percolation with probability T, so the outbreak started by a person is the cluster that contains this person when each edge is kept with probability T. We used 1,000 samples for each T.

The mean-field threshold is $T_c = \langle k \rangle / (\langle k^2 \rangle - \langle k \rangle) = 0.12$. Below it, outbreaks stay tiny. Above it, they grow fast (Fig. 21, left). At $T = 0.2$ an average start reaches 4% of the faculty, and the best start reaches 20%. Who is the best place to start? The k-core predicts it best (Spearman $\rho = 0.93$), better than the degree (0.89), the number of joint papers (0.81) and the betweenness (0.72). This repeats the result of Kitsak and others: the best spreaders are in the *core*, not simply the people with the most links. Our top 20 spreaders are all at KGP, in the materials and chemistry core.

VIII. Connections with Other Indian Institutes

The faculty network stays inside the two IITs. As a last step, we look at co-authorship with all other Indian institutes.

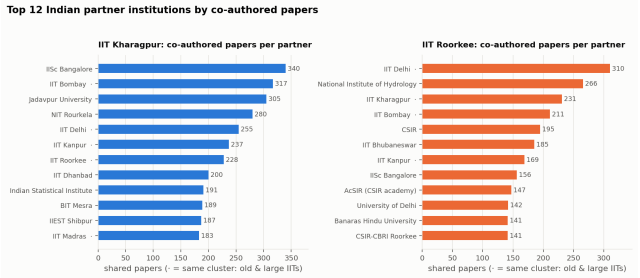


Fig. 22. Top 12 Indian partner institutes of each IIT, by co-authored papers.

A. Partner institutes

We count the shared papers between each IIT and each Indian partner. KGP works most with IISc Bangalore (340 papers), IIT Bombay (317) and Jadavpur University (305). Roorkee works most with IIT Delhi (310), the National Institute of Hydrology (266) and IIT Kharagpur (231) (Fig. 22). These lists make sense by location and by field. The KGP list has several institutes in Kolkata, for example Jadavpur University. The National Institute of Hydrology is in Roorkee, and it matches the strength of Roorkee in water research.

B. Similar or complementary research

Do the IITs work with institutes that do the same research, or with institutes that fill a gap? To measure this, we take the top 20 research subfields of the IIT and of the partner. Then

we calculate the **Jaccard similarity**.

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

The value is 1 if the two lists are the same, and 0 if they have nothing in common.

For the top 25 partners of each IIT, the median is 0.60. Thus most partners do similar research. Some strong partners are complementary. Examples are the Indian Statistical Institute (0.18, KGP), and the National Institute of Hydrology and the Indian Institute of Remote Sensing (0.21 each, Roorkee). These institutes are specialised. Thus the collaboration is probably on one topic, such as statistics, hydrology or remote sensing.

IX. Dashboard and Conclusions

All these results are also in an interactive website, with a switch for KGP, Roorkee or both. A click on a faculty member lists all their papers in our dataset, so anyone can check a person's numbers.

To conclude, both IITs almost doubled their science and engineering output in ten years, and about two thirds of their ranked journal papers are in Q1. Their faculty h-index is about half of a career h-index, because we count only ten years of papers, and it grows by about one point each year. Environmental Science is high because much water and climate engineering is filed there. Papers with more than 10 authors are only 2% of the output, but 6 of KGP's ten most cited papers. Without them the typical numbers stay the same and only the top moves. Citations are heavy-tailed (lognormal-like rather than a clean power law), and new citations go preferentially to papers that are already cited. Most citations inside an institute are self-citations, so we also rank papers without them and correct PageRank for age. Since 2023 Indian research cites the IITs more than they cite it. Faculty work in tight, hierarchical groups inside each campus. The network is a small world that grew denser every year. It survives random loss of faculty, but the loss of about 100 bridging faculty would split the two campuses, and ideas spread best from its core. Each step of this work is a script in our repository, so anyone can repeat it.